

Mobile Transformer Oil Filtration Truck – Upfit

Request for Proposal (Brief Scope) | Delphi Power Equipment Inc. | Houston, TX

1. Project Overview

Delphi Power Equipment Inc. plans to upfit a box truck into a mobile service unit for on-site transformer insulating oil filtration, vacuum oil filling, and hot-oil circulation. The truck will carry its own generator, oil storage, and tools, and must be suitable for all-weather outdoor work at power construction sites.

Cargo box: 161 in × 94.5 in (4,100 × 2,400 mm), interior clear dimensions.

Scope: Turnkey upfit of the box around three Owner-furnished units (below), including design drawings, installation, testing, and documentation. OEM chassis, frame, brakes, steering, suspension, and fuel system must not be cut, drilled, or welded.

2. Owner-Furnished Equipment

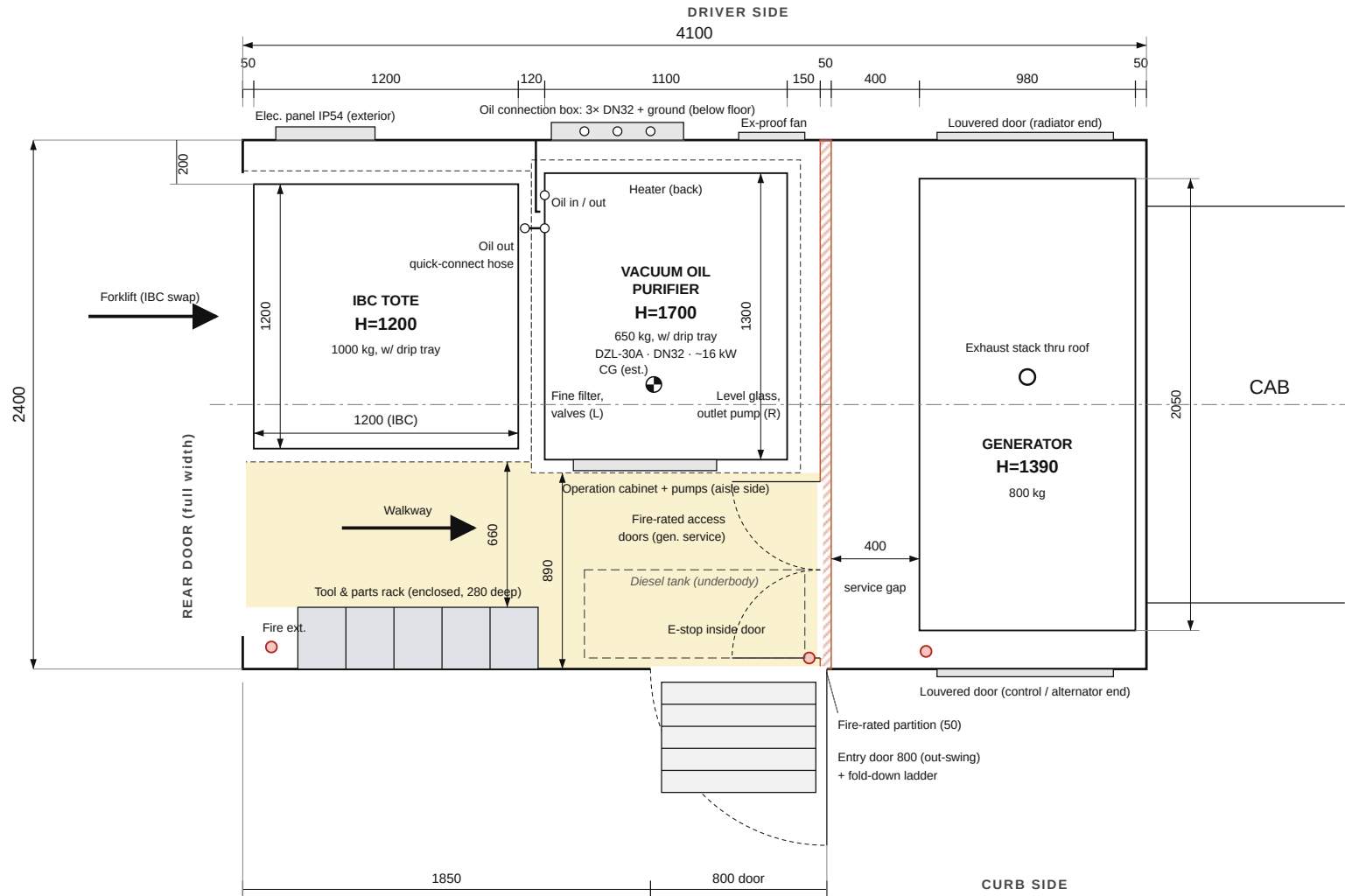
Equipment	Footprint (mm)	Height (mm)	Weight
Diesel generator set	2,050 × 980	1,390	800 kg (approx. 1,760 lb)
Vacuum oil purifier	1,300 × 1,100	1,700	650 kg (approx. 1,430 lb)
Insulating oil IBC tote (full)	1,200 × 1,200	1,200	1,000 kg (approx. 2,200 lb)

3. Proposed Layout

The Owner has selected the layout below. It replaces the two preliminary options in the earlier brief. Please quote on this layout; comments and suggested improvements are welcome. The dimensioned plan is on page 2 and a 3D view on page 3. An interactive 3D model (HTML file, opens in any web browser without installation) is attached.

- **IBC tote** at the rear door, against the driver-side wall; exchanged by forklift through the full-width rear door. Quick-connect hose to the purifier; locating stops and ratchet straps.
- **Vacuum oil purifier** (DZL-30A, approx. 16 kW, DN32 connections) rotated so the 1,100 mm operation side faces the aisle. Casters removed, unit bolted to the channel base, lifting lugs cut after installation. Vacuum pump exhaust piped outside.
- **Generator room** at the cab end (66 kW genset): fire-rated partition with double access doors to the genset service side, 400 mm service gap, louvered service doors at both ends, exhaust stack through the roof with rain cap.
- **Walkways:** main walkway on the curb side 890 mm (35 in); rear walkway 660 mm (26 in).
- **Electrical panel** (IP54) and **oil connection box** (three capped DN32 ports and grounding stud) both on the driver side with exterior access; E-stop inside the side door.
- **Side personnel door** 800 mm, out-swing, on the curb side 1,850 mm from the rear wall, with fold-down ladder. Underbody diesel tank on the curb side; enclosed tool and parts rack on the curb wall.

Proposed Layout – Plan View



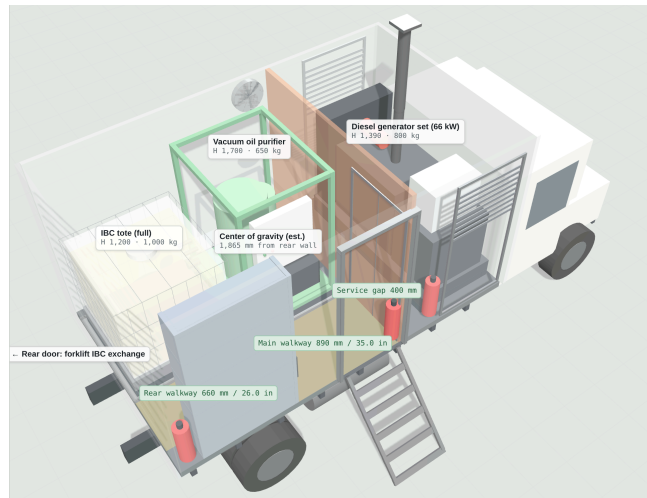
Oil Filtration Truck – Final Layout: IBC at rear door, purifier (operation side to aisle), generator room at cab end

Plan view, cab to the right, mm. Box 4100 x 2400 (161 x 94.5 in). Dashed = drip trays (≥ 2 in rim, 1 in drain). Yellow = walkway: main 890 (35.0 in), rear 660 (26.0 in).

Purifier rotated 90° (1100 operation face to aisle), casters removed, bolted to channel base, lifting lugs cut after install. Vacuum pump exhaust piped outside.

Elec. panel and oil connection box both on driver side, exterior access. Generator: louvered doors both ends, 400 service gap + fire-rated partition doors.

CG (est., incl. rack ~200 kg + full diesel ~200 kg): 1865 from rear wall, 91 toward driver side. Genset 66 kW (purifier ~16 kW). Preliminary – not for fabrication.



3D view of the proposed layout (interactive model attached as HTML)

4. Scope of Work (Summary)

System	Key requirements
Floor & containment	Anti-corrosion coating; oil-resistant anti-slip rubber mat; welded drip pans (≥ 2 in rim) under tote and purifier, each with a 1 in drain valve.
Generator room	Fully enclosed fire-rated, insulated partition at the generator end; rain-proof louvers with forced exhaust fan.
Generator exhaust	Stainless steel exhaust pipe, insulated, routed outside the box away from intakes and oil ports; over-temperature alarm.
Diesel tank	On-board aluminum or stainless tank for the generator, with lockable filler, level gauge, drain, and fuel line.
Oil piping	304 SS main lines plus oil-rated hoses; valving for two modes (tote filtration/filling and closed-loop transformer filtration) with no cross-flow; three capped external ports; 24 h pressure test.
Electrical	IP54 distribution panel with main breaker and protected branch circuits (GFCI); explosion-proof LED lighting and receptacles; oil-resistant cable in conduit.
Grounding	All metal parts bonded; external grounding stud; ground resistance $\leq 4 \Omega$.
Ventilation	Explosion-proof exhaust fan and louvers in the oil work area to remove oil vapor.
Access	Curb-side personnel door per layout (sealed, lockable, with grab handle); fold-down aluminum boarding ladder; interior walkway ≥ 24 in wide.
Cable & tools	Heavy-duty cable reel with power cable; wall-mounted tool rack and enclosed tool storage.
Safety	Equipment on channel-steel bases with anti-vibration mounts and tie-downs; three 4 kg (approx. 9 lb) dry chemical extinguishers; safety signage.

5. Requested from Vendor

- Budgetary price (itemized by the systems above where possible).
- Lead time from approval to delivery, and whether the vehicle needs to be dropped off at your shop.
- Comments on the proposed layout, and any items you recommend adding or changing.
- Similar service / fuel-lube / utility bodies you have built (photos or references).
- Warranty terms.

Detailed technical specifications will be provided to shortlisted vendors.

Contact: Delphi Power Equipment Inc., Houston, TX — Austin@delhipowerequip.com

Reference Photos – Owner-Furnished Equipment and Base Vehicle



Base vehicle



Vacuum oil purifier



Diesel generator set



IBC tote